

面向 IEEE 1838 兼容异构集成系统的电热感知测试访问路径生成与协同调度方法

摘要：

随着 2.5D/3D 异构集成和 Chiplet 系统的发展，测试访问路径呈现出多层次、多通道和强约束并存的特点。IEEE 1838 为三维堆叠集成电路提供了标准化测试访问架构，然而该架构下的 PTAP/STAP 串行级联机制使得测试访问呈现严格的层级依赖，导致传统的独立任务装箱调度策略难以直接适用；在实际测试过程中，PTAP/STAP 串行访问、DWR 分段访问、FPP 并行传输以及 BIST 本地执行往往同时存在，若仅采用固定路径或单一带宽分配策略，容易造成测试时间过长、资源利用率不足以及局部功耗和温度过高等问题。针对上述问题，本文提出一种面向 IEEE 1838 兼容异构集成系统的电热功耗感知测试路径生成与协同调度方法。首先，将芯粒、内核、DWR、STAP、FPP lane、TSV 及中介层互连统一建模为异构测试访问图；随后，提出 Test Access Recipe 模型，将串行访问、并行访问、自测试、混合访问和互连测试抽象为可调度的候选测试配方；在此基础上，引入 CP-SAT 与 ALNS 相结合的混合优化框架，在串行访问互斥、FPP lane 占用、TSV/互连资源、功耗上限、热邻接冲突和 BIST 阶段依赖等约束下，联合优化测试路径选择与测试时序。该方法可避免传统“先定路径、再做调度”带来的局部受限问题，并通过热风险评估与温度反馈机制降低热点叠加风险。本文工作为先进封装场景下 IEEE 1838 兼容测试访问优化提供了一种可扩展的建模与调度思路。

关键词：IEEE 1838；异构集成；测试访问路径；测试调度；电热感知；CP-SAT；ALNS；先进封装

Electro-Thermal-Aware Test Access Path Generation and Co-Scheduling for IEEE 1838-Compliant Heterogeneous Integration

Abstract:

With the development of 2.5D/3D heterogeneous integration and Chiplet systems, test access paths are characterized by multi-hierarchy, multi-channel configuration and stringent constraints. IEEE 1838 provides a standardized test access architecture for three-dimensional stacked integrated circuits. Nevertheless, the serial cascading mechanism of PTAP/STAP in this architecture imposes strict hierarchical dependencies on test access, making conventional independent-task bin-packing scheduling strategies inapplicable directly. In practical testing scenarios, serial PTAP/STAP access, segmented DWR access, parallel FPP transmission and local BIST execution often coexist simultaneously. Adopting fixed paths or a single bandwidth allocation strategy tends to result in excessive test time, low resource utilization, excessive local power consumption and overheating issues. To address the above problems, this paper proposes a thermal-power-aware test path generation and collaborative scheduling method for IEEE 1838-compliant heterogeneous integration systems. Firstly, chiplets, cores, DWRs, STAPs, FPP lanes, TSVs and interposer interconnections are uniformly modeled as a heterogeneous test access graph. Secondly, a Test Access Recipe model is proposed to abstract serial access, parallel access, self-test, mixed access and interconnection test into schedulable candidate test recipes. On this basis, a hybrid optimization framework combining CP-SAT and ALNS is introduced to jointly optimize test path selection and test timing under multiple constraints including mutual exclusion for serial access, FPP lane occupation, TSV/interconnection resource limits, power consumption caps, thermal adjacency conflicts and BIST phase dependencies. The proposed method eliminates local optimization limitations induced by the traditional "path determination prior to scheduling" scheme, and mitigates hotspot superposition risks via thermal risk evaluation and temperature feedback mechanisms. This research provides a scalable modeling and scheduling paradigm for the optimization of IEEE 1838-compliant test access in advanced packaging applications.

Keywords

IEEE 1838; heterogeneous integration; test access path; test scheduling; thermal-power awareness; CP-SAT; ALNS; advanced packaging